

General methods of control of air pollutants

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Harmful Effects of Air Pollutants on Human Health

<i>Pollutant</i>	<i>Harmful effect</i>
Asbestos	Disease of lungs (Fibrosis, asbestosis) tumors, lung cancer
Arsenic	Bronchitis, dermatitis, skin cancer
Aldehydes	Irritation of eyes and respiratory track
Beryllium	Systemic poison causes berilliosis, skin damage
Chlorine	Irritation to eye, nose, throat. Bronchitis, pneumonitis
Cadmium	Highly toxic, causes bronchitis, fibrosis of lungs hypertension, carcinogenic
Lead	Systemic poison, causes anemia (reduction of haemoglobin) affects kidney function, damages brain
Mercury	Highly toxic, enzyme poison, nervous system disorder
Nickel	Respiratory disorder
Pesticides	Various insecticides, herbicides, fungicides, nematicides and repellants are toxic and produce headache, stomach disturbances and irritations.
Radioactive	Somatic (on body cells) effects including leukemia (blood cancer), and other cancers. Genetic effects are mainly mutations in germ cells
Selenium	Irritation to eyes, nose, respiratory tract and gastrointestinal tract, badly affects kidney, lungs and liver
Vanadium	Affects gastrointestinal tract and respiratory tract. Cholesterol synthesis inhibition, carcinogenic

Some of the effective methods to Control Air Pollution are as follows:

1. Source Correction Methods
2. Pollution Control equipment
3. Diffusion of pollutant in air
4. Vegetation
5. Zoning



1. Source Correction Methods

- Industries make a major contribution towards causing air pollution.
- Formation of pollutants can be prevented and their emission can be minimized at the source itself. By carefully investigating the early stages of design and development in industrial processes.
- Example - Those methods which have minimum air pollution potential can be selected to accomplish air-pollution control at source itself.

Types of source correction methods are:

- Substitution of raw materials
- Process Modification
- Modification of Existing Equipment
- Maintenance of Equipment

➤ **Substitution of raw materials:**

- If the use of a particular raw material results in air pollution, then it should be substituted by another purer grade raw material which reduces the formation of pollutants.
 - (a) Low sulphur fuel which has less pollution potential can be used as an alternative to high Sulphur fuels, and,
 - (b) Comparatively more refined liquid petroleum gas (LPG) or liquefied natural gas (LNG) can be used instead of traditional high contaminant fuels such as coal.

➤ **Process Modification:**

- The existing process may be changed by using modified techniques to control emission at source. For example,
 - (a) If coal is washed before pulverization, then fly-ash emissions are considerably reduced.
 - (b) If air intake of boiler furnace is adjusted, then excess Fly-ash emissions at power plants can be reduced.

➤ Modification of Existing Equipment:

Air pollution can be considerably minimised by making suitable modifications in the existing equipment:

- For example, smoke, carbon-monoxide and fumes can be reduced if open hearth furnaces are replaced with controlled basic oxygen furnaces or electric furnaces.
- In petroleum refineries, loss of hydrocarbon vapors from storage tanks due to evaporation, temperature changes or displacement during filling etc. can be reduced by designing the storage tanks with floating roof covers.
- Pressurising the storage tanks in the above case can also give similar results

➤ **Maintenance of Equipment:**

- An appreciable amount of pollution is caused due to poor maintenance of the equipment which includes the leakage around ducts, pipes, valves and pumps etc.
- Emission of pollutants due to negligence can be minimised by a routine checkup of the seals and gaskets.

2. Pollution Control Equipment

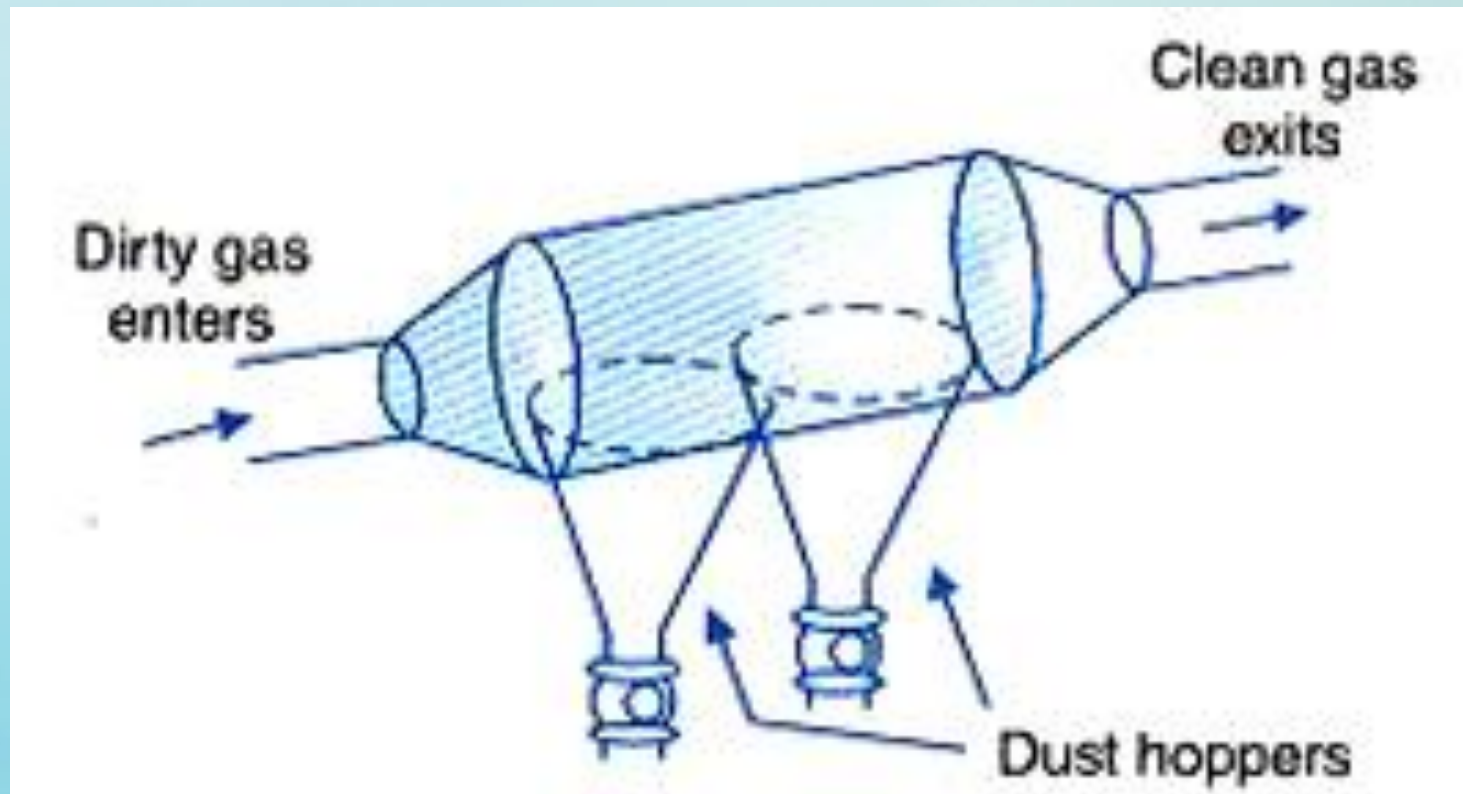


- Control at source is not possible by preventing the emission of pollutants.
- necessary to install them to remove the gaseous pollutants from the gas stream.
- The pollutants are present in high concentration at the source.
- as their distance increases they become diluted by diffusing with environmental air.

These are generally classified into:

- *Gravitational Settling Chamber*
- *Cyclone Separators*
- *Fabric Filters*
- *Electrostatic Precipitators*
- *Wet Collectors*

➤ Gravitational Settling Chamber



■ Mechanism of action

- For removal of particles exceeding $50\text{ }\mu\text{m}$ in size from polluted gas streams.
- This device consists of huge rectangular chambers.
- The gas stream polluted with particulates is allowed to enter from one end.
- The horizontal velocity of the gas stream is kept low (less than 0.3 m/s) in order to give sufficient time for the particles to settle by gravity .
- The particulates having higher density obey Stoke's law and settle at the bottom of the chamber from where they are removed ultimately.
- The several horizontal shelves or trays improve the collection efficiency by shortening the settling path of the particles

❑ Advantages of Settling Chambers

1. Low capital cost.
2. Very low energy cost.
3. few maintenance requirements and low operating costs.
4. Excellent reliability.
5. Dry collection and disposal.

❑ Disadvantages of Settling Chambers

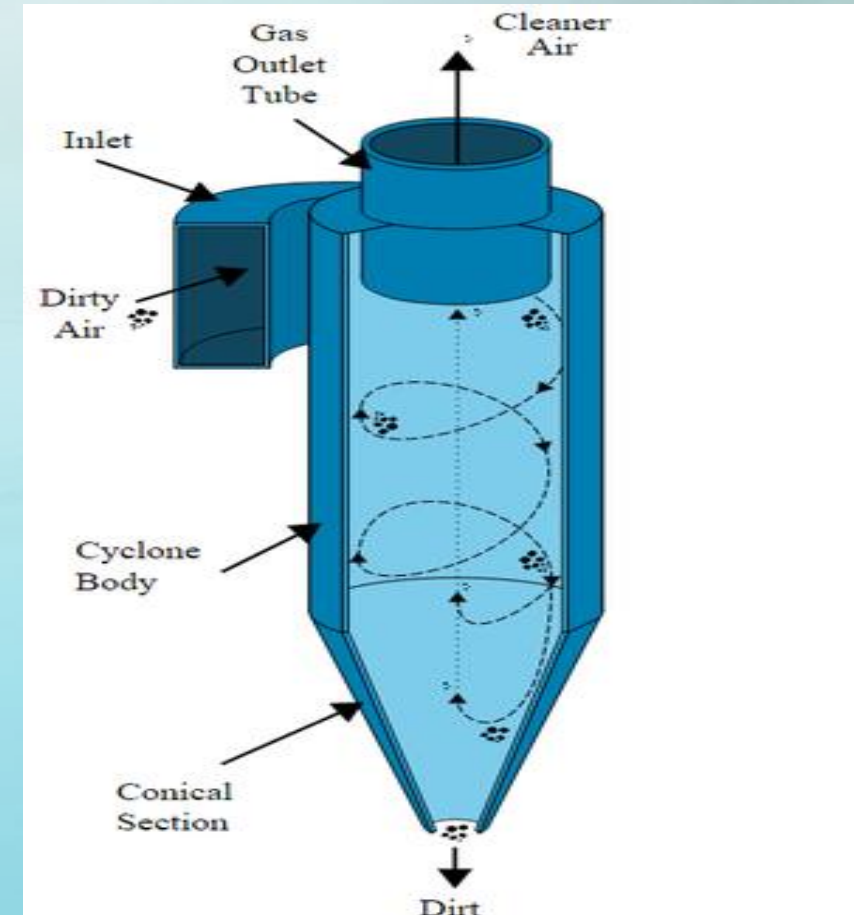
1. Relatively low particulate matter collection efficiencies.
2. Unable to handle sticky or tacky materials.

➤ **Cyclone Separators**

- Centrifugal force is utilized.
- Centrifugal force can be generated by a spinning gas stream
- Can remove smaller particulates.
- It consists of a cylinder with a conical base.
- A tangential inlet discharging near the top and an outlet for discharging the particulates is present at the base of the cone.

■ Mechanism of Action:

- The dust laden gas enters tangentially.
- Receives a rotating motion and generates a centrifugal force
- Due to which the particulates are thrown to the cyclone walls as the gas spirals upwards inside the cone
- The particulates slide down the walls of the cone and are discharged from the outlet.



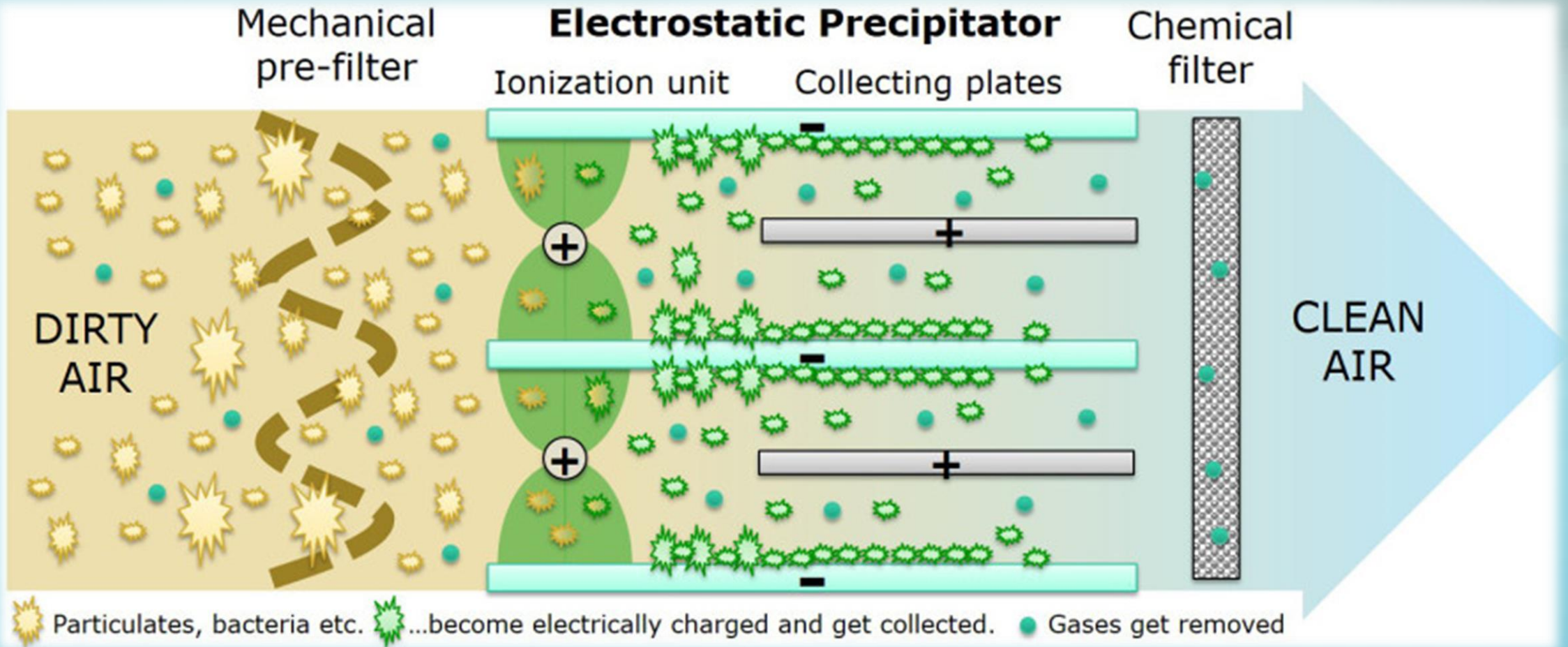
➤ Fabric Filters



■ **Mechanism of Action:**

- Stream of the polluted gas is made to pass through a fabric that filters out the particulate pollutant and allows the clear gas to pass through.
- The particulate matter is left in the form of a thin dust mat on the insides of the bag.
- This dust mat acts as a filtering medium for further removal of particulates increasing the efficiency of the filter bag to sieve 0.5 μm particles.
- a hopper attached at the lower end to collect the particles when they are dislodged from the fabric.
- For efficient filtration and a longer life the filter bags must be cleaned occasionally by a mechanical shaker to prevent too many particulate layers from building up on the inside surfaces of the bag.

➤ Electrostatic Precipitators:



Mechanism of Action:

- It works on the principle of static electricity.
- It removes soot and ash from exhaust fumes before they exit the smokestacks.
- The dirty flue gas escaping through the smokestack is passed through two electrodes.
- One of the electrodes is negatively charged, and this plate causes particulates inside the smoke to obtain a negative charge as they pass by.
- The second electrode is positively charged and thus the negatively charged soot particles are pulled towards it and stick to it.

Advantages:

- Electrostatic precipitation has 99 + percent efficiency
- Can be operated at high temperatures (600°C) and pressure
- Less power requirement
- Economical and simple to operate compared to other devices.

➤ **Wet Collectors**

In wet collectors or scrubbers, the particulate contaminants are removed from the polluted gas stream by incorporating the particulates into liquid droplets.

- **Common wet scrubbers are:**

- (i) Spray Tower

- (ii) Venturi Scrubber

- (iii) Cyclone Scrubber

(i) Spray Tower:

Water is introduced into a spray tower by means of a spray nozzle . As the polluted gas flows upwards, the particulates (size exceeding $10\text{ }\mu\text{m}$) present collide with the water droplets being sprayed downward from the spray nozzles. Under the influence of gravitational force, the liquid droplets containing the particulates settle to the bottom of the spray tower.

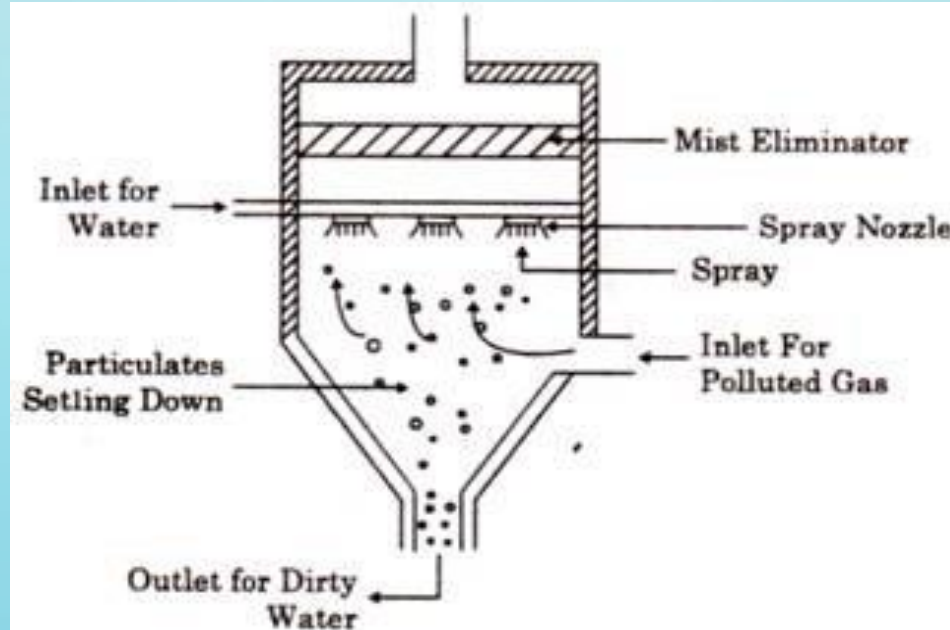


Fig. 5.5. Spray Tower

(ii) Venturi Scrubber :

Submicron particulates (size 0.5 to 5 μm) associated with smoke and fumes are very effectively removed by the highly efficient Venturi Scrubbers. a Venturi Scrubber has a Venturi shaped throat section. The polluted gas passes downwards through the throat at the velocity of 60 to 180 m/sec.

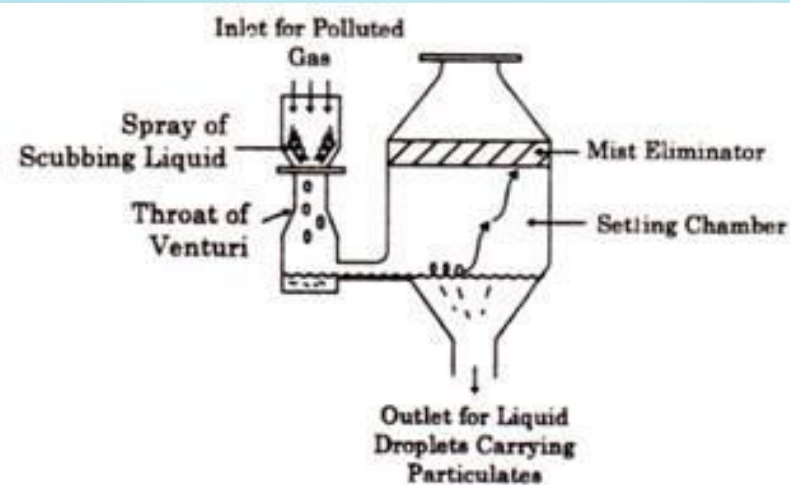


Fig. 5.6. Venturi Scrubber Connected to a Cyclone Separator

Mechanism of action:

- A coarse water stream is injected upwards into the throat where it gets atomized (i.e. breaks the water into droplets) due to the impact of high velocity of the gas.
- The liquid droplets collide with the particulates in the polluted gas stream.
- The particles get entrained in the droplets and fall down to be removed later on.

Advantages:

- Venturi Scrubbers can also remove soluble gaseous contaminants.
- Due to the atomization of water there is proper contact between the liquid and the gas increasing the efficiency of the Venturi Scrubber.

(iii) Cyclone Scrubber :

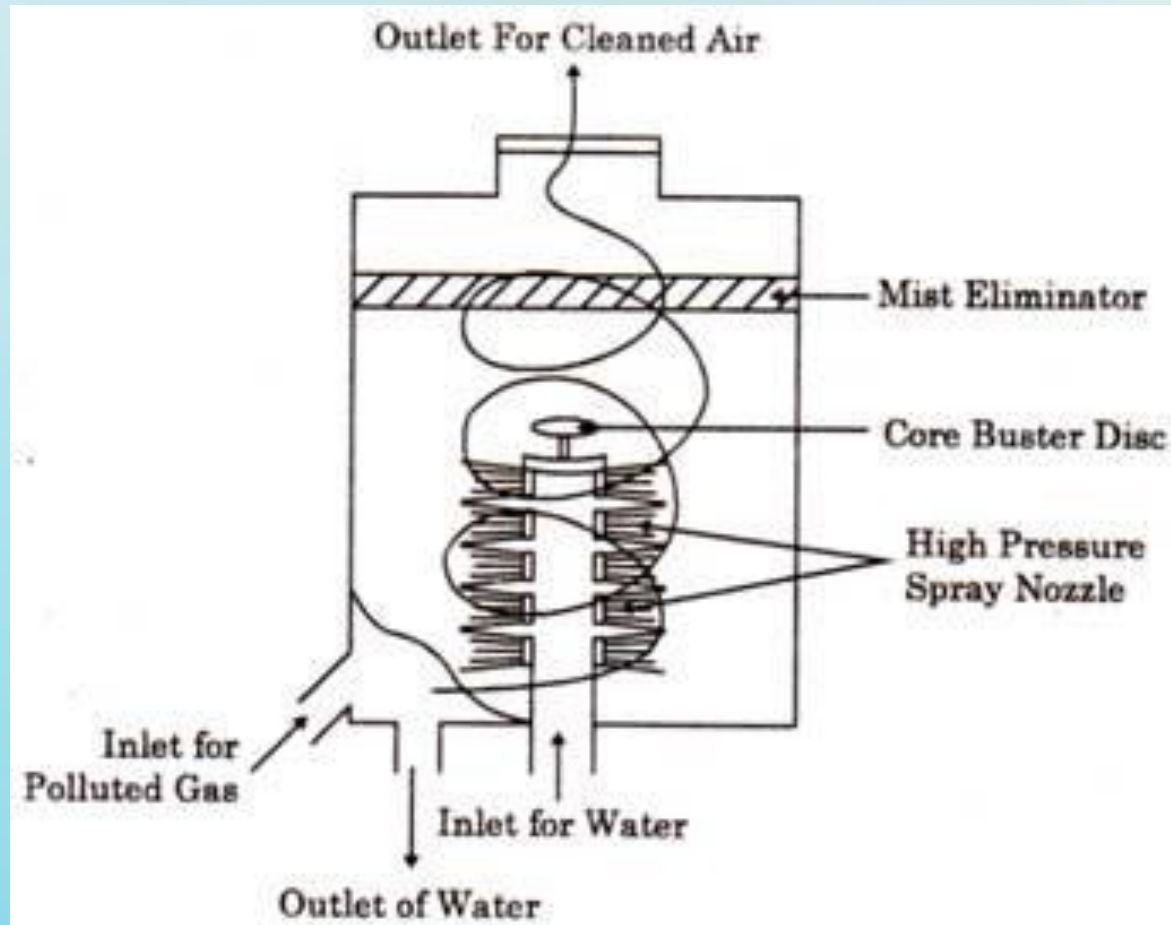


Fig. 5.7. Cyclone Scrubber

Mechanism of action:

- The dry cyclone chamber can be converted into a wet cyclone scrubber by inserting high pressure spray nozzles at various places within the dry chamber .
- The inlet gas enters the chamber tangentially, swirls through the chamber in a corkscrew motion, and exits.
- At the same time, liquid is sprayed inside the chamber.
- As the gas swirls around the chamber, pollutants are removed when they impact on liquid droplets, are thrown to the walls, and washed back down and out.

❑ Advantages of wet scrubbers:

- Can remove both gases and particulate matter simultaneously.
- High removal efficiencies even for submicron sized particulates.
- Valuable by-products like H_2SO_4 can be obtained.

❑ Disadvantages of wet scrubbers:

- High pressure losses
- High operational costs
- Quantity of sludge developed is very high
- Temperature of gas released is decreased. Thus, it will rise to a lower height

3. Diffusion of Pollutants in air

- Tall stacks are used to diffuse pollutants in air.
- Height of stack is 2 to 2.5 times from nearby structures.



Dilution of pollutants in air depends on

- Atmospheric temperature.
- Speed and direction of wind.



Disadvantage of dilution method

It is short term control measure.



Can bring highly undesirable long range effects.

Contaminants come down in another forms at distance.

4. Vegetation

- Trees should be planted in highly polluted area.
- Plants utilizes carbon dioxide and release oxygen.



Plants can fix carbon monoxide

1. *Coleus blumeri*
2. *Ficus variegata*



Some plants can metabolize nitrogen oxides

1. Pinus



2. Pyrus



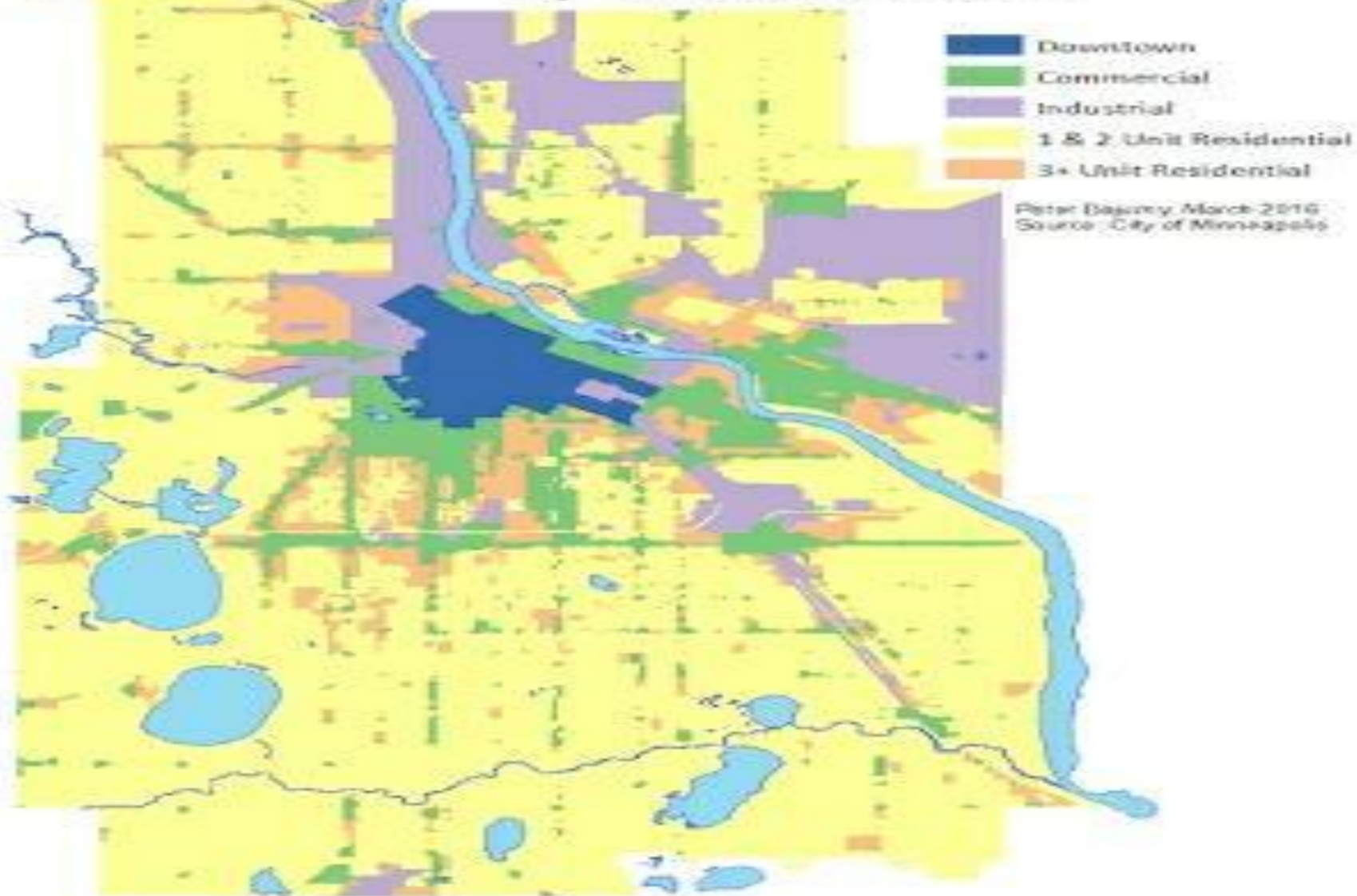
3. Vitis



5. Zoning

- It is a method of controlling air pollution by planning the city.
- Zoning advocates plan the mega cities.
- Heavy industries should not be too close to each other.
- New Industries should be away from city.

Minneapolis Zoning by Classification



Mother earth is choking so are
we!!





Thank
you!!